

Contracts, Residual Cooperation, and Representation in Multi-Agent AI

Q Mechanism Design for Alignment and Control [arXiv:2609.01595](#)

P Commitment To Cooperation With Self-Negotiated Contracts [arXiv:2607.22750](#)

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE

The pair

Q develops a mechanism-design framework for AI agents with uncertain preferences and capabilities, separating feasible reports from incentives to obey after reporting. P studies LLM agents negotiating and using contracts in a sequential bargaining game, and finds that programmatic tile contracts outperform natural-language tile contracts on mutually dependent boards [\[ledger 1, 2, 10\]](#).

The connexion pursued

The initial question was whether P's natural-language contracts expose a failure of Q's assumption that capabilities can be concealed but not counterfeited. That connexion does not hold: Q's restriction concerns evidence supporting *type reports*, whereas P's judge decides whether a proposed move falls within a negotiated agreement. P does not demonstrate counterfeit capability reports. The closer connexion is Q's distinction between feasible actions and obedience, together with its discussion of what a mechanism can measure and enforce [\[ledger 1, 2, 8, 20\]](#).

What was found

P reports that, on mutually dependent boards, both players finish in 46% of natural-language contract games, compared with 79% under programmatic tile contracts and 61% without contracts. The two contract formats cover about 2.6 tiles per game on average, but natural-language contract games have fewer subsequent pay-for-partner proposals and acceptances. Agents declining to trade also cite an existing contract more often. These observations motivate a *residual-cooperation crowd-out* hypothesis: agents may treat an incomplete contract as sufficient and neglect the further exchanges needed to finish. They do not establish that hypothesis causally [\[ledger 2, 3, 15\]](#).

The board permits a sharper diagnostic than average contract size. For player i and accepted tile obligations C , let $\mathcal{P}_i$ be the simple paths from start to goal and define

$$\delta_i(C) = \min_{p \in \mathcal{P}_i} \sum_{v \in p} \mathbf{1}_{\{v \text{ has the opponent's color and } C \text{ does not cover } i \text{ at } v\}}.$$

Each player starts with no chips of the opponent's color, and a contractual coverage pays for its specified move. Any successful route therefore needs at least $\delta_i(C)$ further voluntary opposite-color payments or transfers. In particular, $\delta_i(C) > 0$ certifies that the contract alone cannot get player i to the goal [ledger 14, 17].

The converse fails. A contract may cover every required tile on some individual path yet leave its giver without the chip when the obligation is triggered; the players' chosen paths may also conflict in timing or inventory. An exact certificate that both can finish without further voluntary exchange requires a joint reachability search over positions, inventories, outstanding obligations, and turn order. Even that certificate establishes physical possibility, not that following the route is strategically optimal [ledger 7, 17, 19].

Objections and limits

P's judge audit found three errors among 1,155 approved natural-language contract moves, all false positives. This weighs against frequent erroneous approvals as the explanation, but it says nothing about denied moves or false negatives. Ambiguous clauses may lack a unique correct interpretation until their meaning is fixed independently [ledger 3, 12, 20].

The observed formats also need not contain identical obligations: similar mean coverage does not establish equal path relevance or contractual sufficiency. Q's revelation and peer-discipline results do not predict P's format ranking. Q uses a static mechanism with broadly available rewards, while P has sequential negotiation, inventory constraints, and restricted contract instruments [ledger 6, 8, 10]. A remaining presentation effect under matched obligations and enforcement could reflect planning, beliefs, or equilibrium selection; it would not refute Q's theorem [ledger 12, 15].

Sources and open question

This account uses only Q, P, and the numbered ledger entries. No outside source was consulted. The supplied material lacks the accepted contracts and path-level trajectories needed to determine how often either format left essential cooperation uncovered. It also lacks an audit of denied judge decisions [ledger 12, 15, 20].

Smallest next step

For the existing games, compile each accepted contract into tile obligations and compute $\delta_i(C)$, followed by the joint no-extra-trade reachability flag where possible. Compare later trade requests and completion within these certified groups. To settle whether representation itself causes a difference, replay the same board and fixed obligations with equivalent JSON and mechanically generated prose, keeping enforcement and game information identical; audit both approved and denied moves if judge enforcement is tested separately [ledger 11, 12, 14, 15].